HIDEAKI TAKAHASHI

Ph.D. Researcher in Computer Science (System Security, Formal Verification & AI) | Columbia University

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RESEARCH & PROFESSIONAL SUMMARY

- **Research Focus:** Third-year Ph.D. student in Computer Science at Columbia University specializing in **system security**, **formal verification**, **zero-knowledge proof (ZKP) systems**, and **trustworthy AI / privacy computing**.
- **Publication Track Record:** First-author and co-author of peer-reviewed papers in top-tier (CORE Rank A*) computer security and machine learning conferences, including **ACM CCS 2026**, **IEEE S&P 2026**, **AAMAS 2024**, **ICLR 2024**, and **CVPR 2023**.
- **Real-World Vulnerability Discovery:** Discovered **100+ confirmed zero-day vulnerabilities and CVEs** (earning thousands of dollars in bug bounties) across widely deployed Web3, Zero-Knowledge circuit/zkVM, AI Agent, and Rust ecosystems.
- **Open-Source Leadership:** Creator and maintainer of widely adopted open-source security and formal verification software (**1,500+ total GitHub stars**), including *h5i* (880+ stars) and *AIJack* (400+ stars, 10,000+ downloads).
- **Competitive Machine Learning:** Awarded **5 Kaggle Medals** (2 Silver, 3 Bronze), consistently placing in the top 4%–8% globally across optimization, reinforcement learning, and predictive modeling competitions.

EDUCATION

Ph.D. in Computer Science | **Columbia University** — New York, NY, USA | Sep. 2024 – Present (Expected 2028)

- **Advisor:** Prof. Junfeng Yang | **Affiliation:** Member of The Center for Digital Finance and Technologies.
- **Research Areas:** Automated vulnerability detection, fuzzing and formal verification of complex softwares (Zero-Knowledge circuits, zkVM, agent harness, MCP server, and smart contracts).

B.A.Sc. in Informatics | **The University of Tokyo** — Tokyo, Japan | Apr. 2019 – Mar. 2024

- **Advisor:** Prof. Alex Fukunaga | **GPA:** 3.85 / 4.00 | **Degree:** Bachelor of Arts and Sciences (Informatics).
- **Thesis / Focus:** Privacy-preserving artificial intelligence planning and algorithmic game theory (Transit Obfuscation Problem).

PUBLICATIONS

*Note: * indicates first or equal-first authorship. All conference venues listed below hold CORE Rank A*.*

Peer-Reviewed Conference Papers

[C1] **Hideaki Takahashi**, Suman Jana, and Junfeng Yang. "Efficient Branch-and-Bound Testing and Verification of zkVMs." In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (**ACM CCS 2026**)*, 2026. CORE Rank: A*. [arXiv:2609.15020]

[C2] **Hideaki Takahashi***, Jihwan Kim, Suman Jana, and Junfeng Yang. "zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits." In *Proceedings of the 2026 IEEE Symposium on Security and Privacy (**IEEE S&P 2026**)*, pp. 919–938, May 2026. CORE Rank: A* (Acceptance Rate: 13.0%). [DOI:10.1109/SP63933.2026.00049]

[C3] **Hideaki Takahashi*** and Alex Fukunaga. "On the Transit Obfuscation Problem." In *Proceedings of the 23rd International Conference on Autonomous Agents and Multi-Agent Systems (**AAMAS 2024**)*, 2024. CORE Rank: A* (Acceptance Rate: 20.7%).

[C4] Tianyuan Zou, Zixuan Gu, Yu He, **Hideaki Takahashi**, Yang Liu, Guangnan Ye, and Ya-Qin Zhang. "VFLAIR: A Research Library and Benchmark for Vertical Federated Learning." In *The Twelfth International Conference on Learning Representations (**ICLR 2024**)*, 2024. CORE Rank: A* (Acceptance Rate: 31.1%). [arXiv:2310.09827]

[C5] **Hideaki Takahashi***, Jingjing Liu, and Yang Liu. "Breaching FedMD: Image Recovery via Paired-Logits Inversion Attack." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2023**)*, 2023. CORE Rank: A* (Acceptance Rate: 25.8%). [arxiv:2304.11436]

Preprints & Technical Reports

[P1] Sally Junsong Wang, Jianan Yao, Kexin Pei, **Hideaki Takahashi**, and Junfeng Yang. "Detecting Buggy Contracts via Smart Testing." *arXiv preprint arXiv:2409.04597*, 2024.

[P2] **Hideaki Takahashi***. "AIJack: Security and Privacy Risk Simulator for Machine Learning." *arXiv preprint arXiv:2312.17667*, 2023.

[P3] **Hideaki Takahashi***, Jingjing Liu, and Yang Liu. "Eliminating Label Leakage in Tree-Based Vertical Federated Learning." *arXiv preprint arXiv:2307.10318*, 2023.

[P4] **Hideaki Takahashi***, Kohei Ichikawa, and Keichi Takahashi. "Difficulty of Detecting Overstated Dataset Size in Federated Learning." *IPSJ SIG Technical Reports*, Vol. 2021, No. 10, 2021. [NII ID: 00214220]

RESEARCH EXPERIENCE

Ph.D. Student | **Columbia University** — New York, NY, USA | Sep. 2024 – Present

- Conducting research on automated software testing, fuzzing, and formal verification of cryptographic and blockchain systems under Prof. Junfeng Yang and Prof. Suman Jana [**C1, C2, P1**].
- Formulated the Trace-Constraint Consistency Test (TCCT) and engineered *zkFuzz*, a fuzzing framework that evaluated 452 real-world Circom ZK circuits and uncovered 85 bugs, including 59 zero-day vulnerabilities (39 developer-confirmed, 14 patched) [**C2**].
- Designed efficient branch-and-bound testing and verification techniques for Zero-Knowledge Virtual Machines (zkVMs) [**C1**].

Undergraduate Researcher | **Fukunaga Lab, The University of Tokyo** — Tokyo, Japan | Apr. 2023 – Mar. 2024

- Formulated and solved the Transit Obfuscation Problem (TOP), a novel privacy-preserving AI planning task, under the supervision of Prof. Alex Fukunaga, resulting in a first-author publication at AAMAS 2024 [**C3**].

Research Intern (Federated Learning & Privacy) | **AIR, Tsinghua University** — Beijing, China | Jan. 2022 – Feb. 2023

- Researched adversarial attacks and defenses in federated learning and privacy computing at the Institute for AI Industry Research (AIR) under Prof. Yang Liu and Prof. Jingjing Liu [**C4, C5, P3**].
- Developed the Paired-Logits Inversion Attack demonstrating image reconstruction vulnerabilities in digest-based federated learning (FedMD), published at CVPR 2023 [**C5**], and co-authored the VFLAIR vertical federated learning benchmark library published at ICLR 2024 [**C4**].

Visiting Research Student | **Nara Institute of Science and Technology (NAIST)** Nara, Japan | Aug. 2021 – Sep. 2021
- Investigated free-rider attacks and dataset size falsification in federated learning using MPI, FedML, and PyTorch under Prof. Kohei Ichikawa and Prof. Keichi Takahashi [**P4**].

INDUSTRY EXPERIENCE

Technical Intern: AI/ML & Software Engineer | **Apple Inc.** Yokohama, Japan | Feb. 2024 – Jul. 2024
- Engineered AI/ML and software systems within Apple's machine learning and software engineering organization.

Research Assistant | **UTokyo Economic Consulting Inc.** Tokyo, Japan | Oct. 2020 – Mar. 2024
- Developed applied econometrics and machine learning solutions for industrial and policy consulting engagements, including dynamic product pricing optimization algorithms and geospatial analytics dashboards.

INVITED TALKS

Invited Speaker | **UC Berkeley Security Seminar** Berkeley, CA, USA | Mar. 11, 2026
- **Talk Title:** *zkFuzz: Foundation and Framework for Effective Fuzzing of Zero-Knowledge Circuits.*

Invited Speaker | **Zero Knowledge Frontier Japan: Vol. 2** Tokyo, Japan | Nov. 29, 2025
- **Talk Title:** *Typical Bugs and Detection Methods in Implementing Zero-Knowledge Proof Constraints.*

OPEN-SOURCE SOFTWARE & SECURITY ARTIFACTS

Creator & Lead Maintainer | **h5i (github.com/h5i-dev/h5i)** Rust / Lean 4 | 880+ GitHub Stars
- Engineered an Axum-based Rust web framework specifically architected for formal verification with the Lean 4 theorem prover.

Creator & Lead Maintainer | **AIJack (github.com/Koukyosyumei/AIJack)** C++ / Python | 400+ GitHub Stars
- Developed an open-source security and privacy risk simulator for machine learning (model inversion, membership inference, poisoning, and federated learning defenses), achieving 10,000+ downloads and citations in 8+ external research papers.

HONORS, FELLOWSHIPS & COMPETITIONS

Academic Fellowships & Scholarships
- **Ph.D. Fellowship**, The Columbia-Ethereum Research Center for Blockchain Protocol Design 2025
- **Funai Overseas Scholarship**, The Funai Foundation for Information Technology (Full 2-year tuition & stipend) 2024 – 2026

Machine Learning Competitions (Kaggle — 5 Medals)
- **Silver Medal (45th / 616 teams, Top 8%)** — Kaggle: Google — Fast or Slow? Predict AI Model Runtime 2023
- **Bronze Medal (67th / 875 teams, Top 8%)** — Kaggle: Hungry Geese (Multi-Agent Reinforcement Learning) 2021
- **Bronze Medal (52nd / 788 teams, Top 7%)** — Kaggle: Santa 2020 — The Candy Cane Contest 2021
- **Silver Medal (51st / 1,138 teams, Top 5%)** — Kaggle: Google Research Football with Manchester City F.C. 2020
- **Bronze Medal (88th / 1,390 teams, Top 7%)** — Kaggle: Cornell Birdcall Identification (Audio ML) 2020

ACADEMIC SERVICE
- **Conference Reviewer:** Conference on Neural Information Processing Systems (**NeurIPS 2026**).
- **Journal Reviewer:** *IEEE Transactions on Dependable and Secure Computing (IEEE TDSC)*; *IEEE Transactions on Network Science and Engineering (IEEE TNSE)*; *IEEE Transactions on Medical Imaging (IEEE TMI).*

SELECTED SECURITY FINDINGS
- **CVE-2026-40608** — Out-of-Memory crash in MCP server
- **CVE-2026-42078** — Arbitrary File Write in MCP server
- **CVE-2026-42079** — Arbitrary Code Execution in MCP server
- **CVE-2026-42080** — Arbitrary File Write in MCP server

TECHNICAL SKILLS
- **Programming & Verification Languages:** Rust, Lean 4, C, C++, CUDA, Python, x86/ARM Assembly, LLVM IR, Haskell, Solidity, Circom, Noir, R, Swift.
- **Security & Verification Domains:** Fuzzing, Symbolic Execution, Formal Verification (Lean 4), Zero-Knowledge Proofs (Circom, Noir, AIR, zkVMs), Smart Contract Security, Privacy-Preserving Machine Learning, Federated Learning.
- **AI/ML & Systems Frameworks:** PyTorch, MPI, FedML, OpenMined (PySyft), Axum, Docker, AWS, Google Cloud Platform (GCP).
- **Spoken Languages:** English (Professional Working Proficiency), Japanese (Native).